2 Way ANOVA Example

Ho: average productivity is same under department & Programs

H1: average productivity is same under department & Programs

Level of Significance: α=0.01

Test Statistics

$$F = \frac{MSR}{MSE}$$

$$F = \frac{MSC}{MSE}$$

Critical region

$F > F\alpha, (r-1), (c-1)*(r-1)$➔ $F > F0.01, (3,12)$ $==>$ F>5.95

$F > F\alpha, (c-1), (c-1)*(r-1)$➔ )➔ $F > F0.01, (4,12)$ $==>$ F>5.41

Calculation:

| Program\Dept | Dept 1 | Dept 2 | Dept 3 | Dept 4 | Dept 5 | Ti. | Xij^2 |
|---|---|---|---|---|---|---|---|
| 1 (Workshop) | 12 | 15 | 11 | 14 | 13 | 65 | 855 |
| 2 (Online) | 10 | 12 | 9 | 11 | 10 | 52 | 546 |
| 3 (Hybrid) | 14 | 17 | 13 | 16 | 15 | 75 | 1135 |
| 4 (Coaching) | 16 | 18 | 15 | 17 | 16 | 82 | 1350 |
| T.j | 52 | 62 | 48 | 58 | 54 | 274 | 3886 |

$CF = \frac{T..^2}{n} = \frac{274^2}{20} = 3753.8,$

TSS=$\sum\sum Xij^2 - CF$= 3886-3753.8=132.2, RSS=$\sum \frac{Ti.^2}{c} - CF$=$\frac{65^2+52^2+75^2+82^2}{5} - 3753.8 = 101.8$

CSS=$\sum \frac{T.j^2}{r} - CF$= $\frac{52^2+62^2+48^2+58^2+54^2}{4} - 3753.8 = 29.2$

ESS= TSS-RSS-CSS=132.2-101.8-29.2=1.2

| Source of variation | Df | SS | MS | F |
|---|---|---|---|---|
| RSS | 3 | 101.8 | 33.9333 | 339.333 |
| CSS | 4 | 29.2 | 7.3 | 73 |
| ESS | 12 | 1.2 | 0.1 | |
| Total | 19 | 132.2 | | |

Conclusion: As critical region satisfy therefore there is evidence to reject Ho at 0.01 level of significance we conclude that average production is different under department & programs.